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SCIENCE/TECHNICAL/MANAGEMENT[1]

(a) Scientific Significance and Objectives

In the standard Cold Matter Model (CDM) of cosmology and structure formation, galaxies form when dark matter (DM) density fluctuations grow until they collapse gravitationally to form galactic haloes of dark and baryonic matter, in relative amounts which reflect their average contributions to the cosmic mass density budget. DM dominates the mass and formation of the halo. In fact, when baryons later form stars inside a galactic halo, the energy they release in stellar winds, jets and explosions and ionizing radiation can even reduce their contribution to the galaxy mass budget by blowing interstellar gas out, making the galaxy even more dark-matter-dominated, especially for low-mass galaxies. For this reason, it was very surprising when recent surveys of ultra-diffuse galaxies (UDGs) discovered a new class of galaxies without DM – the so-called "dark matter deficient galaxies" (DMDGs) – challenging the standard CDM paradigm.

Recent surveys of ultra-diffuse galaxies (UDGs) discovered the first two gas-poor UDGs without DM, NGC1052-DF2 (henceforth, "DF2") and NGC1052-DF4 (henceforth, "DF4"), in 2018–19 [64, 66]. Following a flurry of debate regarding the distance determinations of the galaxies [34, 61] – the key observable involved in establishing their DM deficiency – deeper observations by NASA's Hubble Space Telescope have provided new evidence which supports the original claim of anomalously low DM content [5, 53, 65]. In addition, other research has confirmed the existence of DMDGs by discovering other UDGs with enough gas to enable H I atomic gas observations of their kinematics [13, 29–31].

Meanwhile, theoretical attempts (including the FI's) to explain DMDGs within the CDM framework have succeeded in showing that such objects can form as the product of high-speed collisions between galaxies, at hundreds of km/s [**25**, **55**, 56]. This is analogous to the famous Bullet Cluster in which two cluster-sized haloes collided at thousands of km/s and their collisionless components – DM and stars – passed thru each other and separated, while their gaseous baryons, colliding supersonically, were left behind, prevented from interpenetrating by fluid-dynamical behavior involving shocks. Not only does this "mini-bullet" model explain the formation of DMDGs, but it can also explain the exceptionally luminous globular cluster (GC) populations observed [10, 11, 27, 27, 52, 54, 67], by efficient star formation in the dense shock-compressed gas created by the collision [**25**, 62].

The FI and his undergraduate supervisor, Prof. Ji-hoon Kim (collaborator in this proposal) were the first to use high-enough resolution simulations to demonstrate that this mini-bullet model can explain the observed DMDGs within the standard CDM paradigm, including formation of their observed stellar masses and unconventional GC population. In the FI's first paper [**55**] (henceforth, **Paper I**), N-body-hydro simulations by the `ENZO` and `GADGET` codes, with ~ 80 pc length resolution, demonstrated that this mini-bullet model can form DMDGs in the collision center, with stellar mass in line with observed amounts. In the FI's second paper (of which he was lead author) [**25**] (henceforth, **Paper II**), the question of GC formation was addressed, by higher-res `ENZO` galaxy-collision simulations (1.25 pc resolution), as required to model this process in sufficient detail, succeeding in forming the observed massive GC population, as well (see Fig. 1).

Other suggested possible explanations include the tidal dwarf galaxy (TDG) scenario [2, 37], in which dwarf galaxies formed out of tidally-stripped gas removed from galactic

[1] **NOTE: Cited papers by the FI (Joohyun Lee), PI (Paul Shapiro), or collaborator (Ji-hoon Kim) are highlighted in bold-face throughout.**

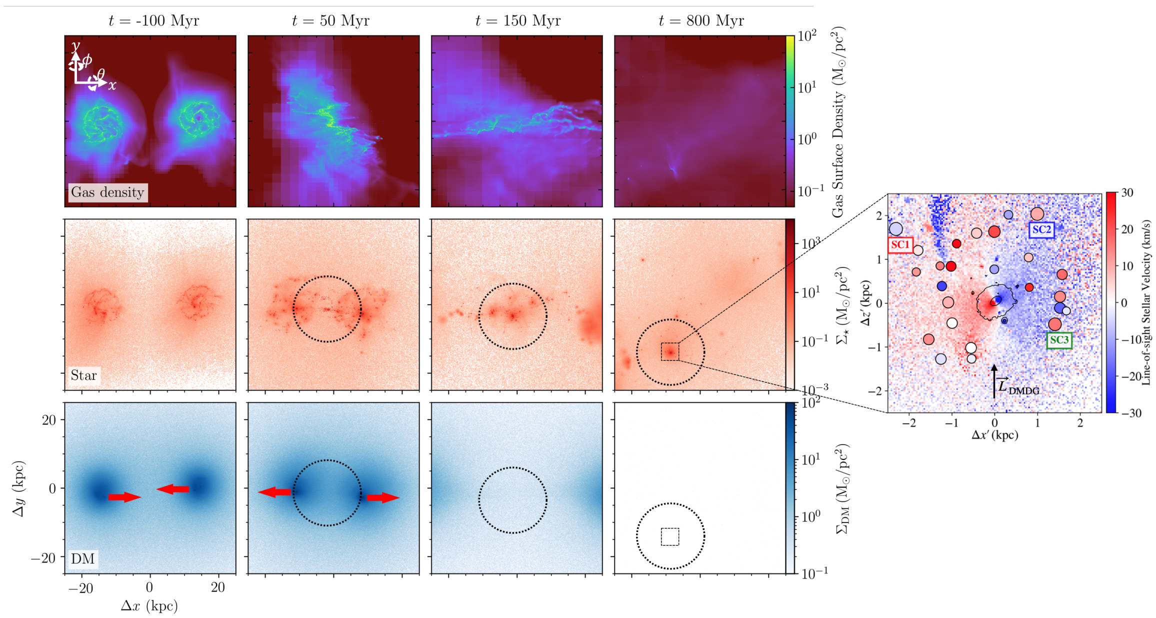

Figure 1: Illustration of the simulated “mini-bullet” scenario taken from the FI’s **Paper II**. Surface densities of gas (top row), stars (middle row) and DM (bottom row) are shown. Two progenitor galaxies collide with each other to separate baryons from DM invoke a severe shocks in ISM ($t = -50$ Myr). Following the shock compression, ISM efficiently cools and clumps to form massive SCs with $M_\star \sim 10^5 - 10^6\,\mathrm{M}_\odot$ (t = 150 Myr). 800 Myr after pericentric approach of the progenitors, 42 member SCs are orbiting the DMDG with $M_\star \sim 3 \times 10^8\,\mathrm{M}_\odot$ (in the dotted circles of 10 kpc radius; t = 800 Myr). A zoom-in plot of the rectangle area indicating the location of three most massive SCs (SC1, SC2, SC3) and other SCs with their line-of-sight velocity viewed from a fixed axis is depicted (rightmost panel).

subhaloes, and the tidal stripping of DM from such subhaloes [8, 15, 16, 28, 35, 38, 41–43, 68]; however, recent observational results cast doubt on those scenarios, by confirming there are no distinct tidal features in the observed DMDGs [18, 36]. Moreover, these other suggestions fail to account for the massive GCs.

Some additional effort has been devoted to searching for DMDGs in the halo catalogues of pre-existing large-scale cosmological simulations of galaxy formation in ΛCDM, which found them to be rare enough to question the compatibility of DMDG observations with the current state of such simulations [14]. However, such large-scale cosmological simulations may lack the ability to form DMDGs. For example, they do not have sufficient spatial resolution to resolve the details of the high-speed galaxy collisions and shocks involved in the mini-bullet model, and are forced to rely on subgrid algorithms for star formation which were calibrated to match observations of ordinary galaxies. Those star formation algorithms may fail within the highly shock-compressed gas that occurs in such high-speed collisions. While there are also higher-res zoom-in simulations in selected, small subvolumes of large-volume cosmological simulations, these zoom-in regions, to-date, were not selected to contain colliding galaxy pairs with high relative velocity, so no previous work using cosmological simulations could have resolved the mini-bullet.

As new observations accumulate additional details, our previous proof-of-principle simulations must now be followed-up by new high-res, cosmological N-body and gas dynamics simulations to match them. Advances will include: (1) a higher degree of physical realism by improved treatment of radiative processes and subgrid algorithms for stellar physics; (2) larger simulation volumes, to follow each collision history more fully, from widely-separated initial halos to the final stage of widely-separated, dark-matter-dominated halos, far from the central DMDGs they leave behind; (3) improved initial conditions (ICs) to better match the latest DMDG observations, by refined choice of colliding halo properties and their self-consistent emergence during cosmological structure formation; (4) to determine the statistical likelihood of occurrence of these ICs in the ΛCDM universe, large volume, dark-matter-only simulation halo catalogues already available will be analyzed. Our results will test the CDM paradigm by determining both the collision parameters capable of reproducing observed DMDGs and their statistical likelihood, while predicting what future surveys should detect.

The mini-bullet model works to explain DMDGs because of the difference between the collisionless nature of CDM and fluid-like behavior of gaseous baryons. DMDGs, therefore, present a unique opportunity to test and constrain alternative DM models with fluid-like behavior on small scales, such as self-interacting DM (SIDM). Just as the Bullet Cluster provides a fundamental upper limit to the SIDM particle scattering cross section, we propose now to use DMDGs to push this constraint to the lower energy of particle collisions in mini-bullets. This will be done by replacing the N-body code used above for CDM, by one which adds a Monte-Carlo algorithm for SIDM elastic scattering.

This research serves NASA's strategic objectives, by advancing our understanding of the origin and evolution of galaxies and structure in the universe, as well as the nature of DM. Its goals and outcomes will contribute to two focused programs, Physics of the Cosmos program and Cosmic Origins program, of the Astrophysics Research Program within the Astrophysics Division of NASA's SMD.

(b) Proposed Research and Methodology

1. Simultaneous Formation of Dark Matter Deficient Galaxies and Star Clusters by High-speed Galaxy Collision

The first suggestion of the "mini-bullet" scenario for DMDG formation involved a high-velocity collision of two gas-rich, dwarf-sized galaxies at ~ 300 km/s [56]. It was not a detailed calculation or simulation, however. The first attempt to simulate this formation scenario [**Paper I**] in detail was by the FI and his supervisor, as described in **Paper I** and **Paper II**, using two different codes, a particle-based SPH code `GADGET-2` [59], with Tree-Particle-Mesh (TPM) N-body solver, and second, to increase spatial resolution, a grid-based, Eulerian, Adaptive Mesh Refinement (AMR) code, `ENZO` [**3**]. In **Paper I**, a suite of simulations, using both codes, found the allowed range of ICs for the colliding galaxies (e.g. their initial velocity magnitudes and directions, total masses and mass ratios, and initial mass fractions in stars and gas) needed to form DMDGs with $M_\star \sim 10^8\,\mathrm{M}_\odot$, post-collision. These requirements were that colliding progenitor galaxies with $M_{\mathrm{total}} \sim 5 \times 10^9\,\mathrm{M}_\odot$ should contain enough gas, $M_{\mathrm{gas}} \gtrsim$ few $\times\, 10^8\,\mathrm{M}_\odot$, collide with high velocity of $\sim 200 - 800$ km/s, and at exceptionally small impact parameter of only a few kpc. In addition, when stars and DM separated, post-collision, from the gas, some collision parameters produced multiple DMDGs, up to 5, depending on the initial velocity and gas masses involved – a prediction which new observations (private communication, P. van Dokkum) appear to confirm.

During the collision, when the collisionless components – DM and stars – passed thru each other and separated as gas-free DM-dominated haloes, the ISM of each galaxy was prevented from doing so by a pair of high-Mach-number radiative shocks which decelerated the approaching supersonic gas streams, between which the gas rapidly cooled and compressed by factors of order the square of the Mach number, leading to gravitational instability and the possibility of star formation. While these **Paper I** simulations were not highly-enough resolved to follow this in enough detail, results were reminiscent of the radiative shock model for GC formation inside protogalaxies of the PI and his collaborators [**17**, **49**, **51**]. This justified the hope that, if the simulation could be repeated with higher resolution, it might reveal the formation of the massive GCs discovered by DMDG observations. That was the motivation for the higher-res `ENZO` simulations of **Paper II** by the FI.

At **Paper II**'s higher resolution, subgrid star formation and feedback algorithms also had to be revised, to account for smaller grid-cells. The new simulations using the same ICs as **Paper II** confirmed the formation of DMDGs (see Fig. 1). While the star clusters (SCs) had a mass distribution in line with the observed GCs, there is a discrepancy in the half-mass (light) radii and metallicities ([M/H]) due to limited spatial resolution which may also lead to excessive self-enrichment of star forming gas clumps.

2. New Suites of Galaxy Collision Simulations

While the previous studies [**Paper II**, **Paper I**] supported the mini-bullet scenario as a promising formation channel for the observed DMDGs by using two distinct simulation codes and numerical resolutions, the micro-physics and subgrid algorithms relevant to SC formation and gas heating/cooling can be improved, including better treatment of stellar feedback, supermassive black hole (SMBH) feedback, and radiative transfer (RT). One of the caveats of our previous work is a discrepancy between half-light radii (r_h) of DMDGs formed in the simulations and the observed values. Stellar feedback is treated as a pure supernova (SN) thermal energy injection, which can, however, involve other kinds of feedback, such as early stellar wind, radiative feedback, and momentum feedback. More realistic stellar feedback that is able to efficiently push nearby ISM, making the stellar component of the DMDGs more expanded, may fill the gap between the morphology of DMDGs in the simulation and the observation [63]. Another caveat was that the simulations did not include AGN (SMBH) feedback. One of the signatures that the mini-bullet model predicts is the absence of SMBHs in the collision-induced DMDGs, since SMBHs, expected to be present in the pre-collision galaxies, should continue onward post-collision, not bound to the DMDGs, just as DM haloes and pre-collision stars do. **Our objectives are (1)** to find structural and collisional parameter ICs that will produce collision outcomes that match new DMDG observations, from a suite of low-cost intermediate-res `ENZO` sims; **(2)** to reproduce new DMDG observations using these refined ICs by running higher-res `ENZO` sims in larger volumes, for full-collision-history study; and **(3)** to account for the effect of radiative feedback from ionizing starlight, throughout this history, on formation of DMDGs and their massive SCs.

For **(1)** and **(2)**, by using `ENZO` again, we can use our existing pipelines for generating ICs and analysis of simulation outcomes. In addition, our collaborator, Prof. Ji-hoon Kim, will contribute his expertise as a co-developer of the code. Prof. Kim, who coauthored **Paper I** and **Paper II**, developed `ENZO`'s subgrid algorithm for AGN feedback by SMBH accretion [**19**]. Our main goal is to find parameters that reproduce the recent observational results for DF2, DF4, and other nearby UDGs from van Dokkum (private communication) – to test the hypothesis that they all emerge as the multiple DMDGs produced in a single collision. van Dokkum estimates the stellar mass of DF2 and DF4 ($\sim 10^8\,\mathrm{M}_\odot$) and number of DMDGs ($\sim 8$), along with their age ($\sim 6-9$ Gyr) and relative velocity (~ 350 km/s). We will find the allowed range of collision parameters for our high-res sims by performing a large suite of intermediate-res sims with `ENZO`, that are much less costly.These runs, with few-pc resolution, will adopt various ICs, including both disk galaxies and dwarf spheroidals for the first time. With these ICs derived as objective **(1)**, we will perform the higher-res ENZO sims of objective **(2)**.

With these same ICs, we will run similarly high-res sims to achieve objective **(3)**, by using the RAMSES-RT code. This choice reflects the importance of accounting for feedback from hydrodynamical back-reaction of photoionization by UV starlight from massive stars, including GCs that form. RAMSES-RT is a grid-based, AMR radiation-hydro code with an M1 closure moment-based treatment of RT and non-equilibrium ionization of H and He [48]. As shown by the PI's Cosmic Dawn (CoDa) simulations of galaxy formation during cosmic reionization [**7, 39, 40**], such feedback is especially important in small-mass halos ($M_{\rm halo} \lesssim 10^9\,{\rm M}_\odot$), for which it can lower their gas supply and suppress star formation, in a mass range that overlaps that of DMDGs. Hence, including RT in our sims may affect the DMDGs that result, significantly. The PI has extensive experience with RAMSES, especially its radiation-hydro version, RAMSES-CUDATON, used in CoDa simulations. His CoDa team also includes co-developers of both RAMSES-CUDATON and RAMSES-RT.

All simulations will be run with current and future supercomputer resources of the PI: Frontera and Stampede2 at the Texas Advanced Computing Center (TACC), through local UT Austin allocations, and NSF-funded XSEDE allocations on Stampede2 at TACC and Bridges2 at the Pittsburgh Supercomputing Center (PSC), and Oak Ridge Leadership Computing Facilities (OLCF) through his DOE INCITE grant.

3. Statistical Likelihood of The Galaxy Collision in the ΛCDM Universe

While detailed modelling of mini-bullet galaxy collisions is necessary to test the model against DMDG observations, the ICs thereby found to be required must then be of high enough probability in the ΛCDM universe to explain the frequency of occurrence of DMDGs. The first attempts to interpret observations of the analogous case of the Bullet Cluster in terms of the high-speed head-on collision of merging sub-clusters, to establish their ICs, claimed so high a collision velocity was required that the event was extremely unlikely in ΛCDM – a strong challenge to the ΛCDM paradigm [e.g.24]. It turned out, however, to be necessary to simulate, not only gravitational dynamics, but gas dynamics, too, so as to match all the observations (including X-rays from shock-heated gas) properly, with high enough resolution, in order to determine the collision velocity more accurately. Once the PI and his co-authors in [**33**] did this, they found a lower velocity than earlier work, which dramatically increased the statistical likelihood that such an event could happen in ΛCDM, moving away from the cliff-hanger problem it was previously thought to be for ΛCDM (e.g. [23, 32, 60]). This example is a perfect analog that shows why we are proposing to simulate the mini-bullet model to match observations of the DMDGs, to determine what initial conditions are required to reproduce them.

To decide the statistical probability of finding the mini-bullet galaxy collision in the ΛCDM Universe, we will use two sorts of large-scale cosmological simulations. First, we will find the best publicly-available, large-volume ($\sim (1\,{\rm Gpc}/h)^3$), DM-only simulations capable of addressing the formation of collision progenitors of DMDGs, with its halo catalogue (e.g. the Farpoint simulation [12]). Using the best choice, we will analyze its catalogue, to find halo pairs that closely match relative velocity, masses, mass ratio, separation distance, and collision angle of the ICs required. Such a large box size is necessary to provide good statistics for the likelihood of such rare galaxy collisions. The second simulations are small-volume simulations with ICs from the Constrained Local Universe Simulations (CLUES) project [4, 57], as used by the PI's CoDa collaboration, which includes CLUES members. The CLUES project generated cosmological ICs constrained from the observed main features in the local universe, such as the Local Supercluster and Virgo cluster. The simulation volume is $(64\,\mathrm{Mpc}/h)^3$, which is enough to encompass the distance to the observed DMDGs (20 Mpc for DF2 and DF4).

4. Testing Alternative Dark Matter Models as a Substitute of Cold DM

The mini-bullet model succeeds in explaining DMDGs because of the difference between the collisionless nature of CDM and fluid-like behavior of gaseous baryons. DMDGs, therefore, present a unique opportunity to test and constrain alternative DM models. The PI has extensive experience in study of structure formation in such alternative DM models which depart from CDM by introducing collisionality (e.g. SIDM, [**1**, **21**, **22**]) or fluid-like behavior (e.g. SFDM, [**6**, **26**, **44**, **50**]) on small scales; we will focus on the first of these, namely SIDM [9, 58]. The Bullet Cluster provides a fundamental upper limit to the SIDM particle scattering cross section [20, 45]. By extension, **we propose to use DMDGs to push this constraint to the lower energy of particle collisions in mini-bullets.** This will be done by replacing the N-body code used above for CDM, by the `GIZMO` code with its SIDM module, with a Monte-Carlo algorithm for SIDM elastic scattering [46, 47].

(c) Anticipated Accomplishments and Major Milestones

Year 0 – Pre-fellowship preparation (01/2022 - 08/2022): Large suite of low-cost, intermediate-res (few pc) simulations using `ENZO`, to delineate the range of allowed ICs for galaxy collisions that reproduce the latest DMDG observations. (see Section (b) 2.).

Year 1 (09/2022 - 08/2023): Using these newly refined ICs from Year 0, run high-res (pc-scale), large-box (5 Mpc, i.e. full-collision-history tracking volume) `ENZO` simulations, like those the FI produced in **Paper II** but now in larger volume, to match physical properties of observed DMDGs and GCs, following the evolution to the late stage when initial progenitor haloes are gas-poor, DM-dominated haloes, and widely separated, post-collision.

• Publish **Paper 1** presenting the formation of DMDGs and GCs that match existing and predict new observations (see Section (b) 2.).

Year 2 (09/2023 - 08/2024): Using refined ICs produced in Year 0, run high-res rad-hydro simulations of the same large-box as Year 1, but using `RAMSES-RT` code to study the additional effect of back-reaction of stellar feedback on ISM and IGM gas, both from stars inside the galaxies and from external sources during the progenitor's pre-collision beginnings at higher redshift, including reionization.

• Publish **Papers 2 and 3**. Paper 2 will describe our treatment of radiative transfer and radiative feedback, and present the detailed results of the simulations. Paper 3 will apply these results to predict observations of DMDGs (see Section (b) 2.).

Year 3 (09/2024 - 08/2025): Having successfully matched observed DMDGs with galaxy collisions in the allowed range of collision parameters by simulations above, test their likelihood in LCDM by analyzing existing, large-volume cosmological simulations to derive the statistical likelihood of these required galaxy collision ICs in the ΛCDM universe.

• Publish **Paper 4**. Discuss how rare such high-velocity collisions are and if ΛCDM is challenged (see Section (b) 3.).

Model mini-bullets in the SIDM model, employing the procedures we used in previous years to derive ICs and run simulations, replacing CDM with SIDM by using the `GIZMO` code modified to include Monte-Carlo SIDM scattering algorithm (see Section (b) 4).

• Publish **Paper 5**. Compare SIDM simulation outcomes with the DMDG observations, to impose a new upper limit on the SIDM particle scattering cross section at the lower energies of particle collisions in mini-bullets as compared to those in the Bullet Cluster (see Section (b) 4.).

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RESEARCH READINESS STATEMENT

I have been deepening my understanding of the theory of cosmological large-scale structure and galaxy formation and evolution, and computational astrophysics/cosmology throughout my undergraduate and graduate degree programs and research experiences. I took graduate level courses in astrophysics, in my undergraduate (Physics) and graduate (Astronomy) programs, receiving a grade of A in all the following: classical mechanics, extragalactic astronomy, interstellar medium (ISM), gas dynamics, radiative process, and cosmology. This semester, my classes are in the theory and methodologies of gravitational dynamics and computational astrophysics. By these courses, I gained a deeper knowledge about the current paradigm of cosmology and galaxy formation, physical processes governing the ISM and IGM, including how gas/stars behave under gravitational and hydrodynamic interactions, and how to model such physics in analytical and computational research. As a double major, in Physics and in Electrical and Computer Engineering (ECE), ECE classes in computer organization, data structure, and programming methods taught me to use diverse programming languages, including `C`, `C++`, `Python`, and `Linux` operating system.

In addition, I actively participated in summer/winter schools hosted by various institutes. In Korea Institute for Advanced Study Astrophysics schools, I learned how real research uses the theories I learned in courses and what problems the pioneering field of astrophysics is trying to address. I gained experience with computational methods commonly used in astrophysics, including machine learning (ML), big data analysis, and gravitational and gas dynamics codes. For example, I learned about gravitational N-body codes, by both lectures and a project using the Tree-Particle Mesh (TPM) code. I also learned the ML decision-tree algorithm and used it to analyze a galaxy catalogue from large-scale cosmological `RAMSES` simulation `Horizon Run 5` and study correlations amongst galaxy properties.

These learning experiences prepared me well to undertake my first research projects, an observational one on co-evolution of AGNs and host galaxies, and the one with my undergraduate supervisor, Prof. Ji-hoon Kim (proposal collaborator), described in the Science/Technical/Management section, which resulted in my first two publications (see Paper I [**55**] and II [**25**], the latter, in *ApJL*, of which I was lead author). This involved two kinds of simulations of high-speed collisions of two galaxies to model the formation of newly-discovered dark-matter-deficient galaxies (DMDGs) — one using the `GADGET` cosmological gas and N-body code, a Lagrangian (particle-based) code, and the other, an Eulerian (grid-based) code with Adaptive Mesh Refinement (AMR), `ENZO`, (which I will use again for the proposed research) – both MPI-massively-paralleled codes running on supercomputers. I gained additional experience and improved my communication skills in English, by giving several talks on this work, at local and international meetings.

Thank to this previous research experience, I am now capable of running, modifying and writing simulation and analysis codes in `C`, `C++`, `Python`, and `Fortran`, essential to conducting the proposed research. I am fortunate to be supervised by Prof. Paul Shapiro (PI), an expert on cosmology, structure formation, gas and N-body dynamical simulation, and the nature and origin of dark matter, involving the theory and some of the codes for my proposed research. Through him, I have abundant access to the supercomputers of the NSF-funded TACC supercomputer center here at UT, and access to other centers, as well. In addition, my former supervisor, Prof. Kim, will collaborate with us on this research.

Graduate Study Timeline

Degree type: Ph.D. Subject area: Theoretical and Computational Astrophysics
Time since enrollment: 8 months (since 08/2021) Estimated graduation date: 08/2027

Curriculum Vitae of PI: Paul R. Shapiro

Professional Preparation

PhD: Harvard University, Astronomy, 1979

A.B. (Summa cum Laude): Harvard College, Astronomy, 1974

Appointments

2006 – present, Frank N. Edmonds, Jr. Regents Professor in Astronomy.

1992 – present, Professor of Astronomy, The University of Texas at Austin.

1986 – 1992, Associate Professor; 1981 – 1986, Assistant Professor.

1978 – 1980, Postdoctoral Member, The Institute for Advanced Study, Princeton.

Awards and Fellowships

Elected to Chair-line, American Physical Society Division of Astrophysics, 2013-2017; Fellow of American Physical Society 2011- ; Frank N. Edmonds, Jr. Regents Professor in Astronomy, 2006- ; John W. Cox Fellow for Advanced Study in Astronomy, U. Texas, 2004-2006; National Chair of Excellence, UNAM (Mexico), 1997; Big XII Fellowship, U. Kansas, 1997; Dean's Fellowship, U. Texas, 1997; Alfred P. Sloan Research Fellowship in Physics, 1984-88; Astronomy Teaching Excellence Award, U. Texas, 1988; NSF Predoc. Fellowship in Astronomy, 1974-77; Parker Fellowship, Harvard, 1977-78; Smithsonian Predoc. Fellowship, 1977-78; Phi Beta Kappa, 1974; Sigma Xi, 1974; Harvard Book Prize, 1970; Rensselaer Math and Science Medal, 1970; Math. Assoc. of America Bronze Pin, 1970; National Merit Scholarship, 1970.

Selected Publications Most Relevant to This Proposal

1. Photoevaporation of Cosmological Minihalos During Reionization, **Shapiro, P. R.**, Iliev, I. T., Raga, A. C. 2004, *MNRAS*, **348**, 753. (astro-ph/0307266)
2. Formation and Evolution of the Self-Interacting Dark Matter Halos, Ahn, K., **Shapiro, P. R.** 2005, *MNRAS*, **363**, 1092. (astro-ph/0412169)
3. C^2-Ray: A New Method for Photon-conserving Transport of Ionizing Radiation, Mellema, G., Iliev, I. T., Alvarez, M. A., **Shapiro, P. R.** 2006, *NewA*, **11**, 374–395. (astro-ph/0508416)
4. Simulating Cosmic Reionization at Large Scales I: the Geometry of Reionization, Iliev, I. T., Mellema, G., Pen, U. L., Merz, H., **Shapiro, P. R.**, Alvarez, M. A. 2006, *MNRAS*, **369**, 1625. (astro-ph/0512187)
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6. The Cluster-Merger Shock in 1E 0657-56: Faster than the Speeding Bullet?, Milosavljević, M., Koda, J, Nagai, D., Nakar, E, **Shapiro, P. R.** 2007, *ApJL*, **661**, L131. (astro-ph/0703199)
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8. Detecting the Rise and Fall of the First Stars by Their Impact on Cosmic Reionization, Ahn, K., Iliev, I.T., **Shapiro, P. R.**, Mellema, G., Mao, Y., Koda, J., 2012, *ApJL*, **756**, L16. (arXiv:1206.5007)
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11. Cosmic Dawn (CoDa): the First Radiation-Hydrodynamics Simulation of Reionization and Galaxy Formation in the Local Universe, Ocvirk, P., Gillet, N., **Shapiro, P. R.**, ..., 2016, *MNRAS*, **463**, 1462. (arXiv:1511.00011)

12. The Hydrodynamic Feedback of Cosmic Reionization on Small-Scale Structures and Its Impact on Photon Consumption during the Epoch of Reionization, Park, H., **Shapiro, P. R.**, ..., 2016, *ApJ*, **831**, 86. (arXiv:1602.06472)
13. The Inhomogeneous Reionization Times of Present-day Galaxies, Aubert, D., Deparis, N., Ocvirk, P., **Shapiro, P. R.**, Iliev, I. T., Yepes, G., Gottlöber, S.; Hoffman, Y., Teyssier, R., 2018, *ApJL*, **856**, L22. (arXiv:1802.01613)
14. Suppression of Star Formation in Low-Mass Galaxies Caused by the Reionization of their Local Neighborhood, Dawoodbhoy, T., **Shapiro, P. R.**, ..., 2018, *MNRAS*, **480**, 1740. (arXiv:1805.05358)
15. Cosmic Dawn II (CoDa II): A New Radiation-Hydrodynamics Simulation of the Self-Consistent Coupling of Galaxy Formation and Reionization, Ocvirk, P., Aubert, D., Sorce, J. G., **Shapiro, P. R.**, ..., 2020, *MNRAS*, **496**, 4087 (arXiv:1811.11192)
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17. The Impact of Inhomogeneous Subgrid Clumping On Cosmic Reionization, Mao, Y., Koda, J., **Shapiro, P. R.**, ..., 2020, *MNRAS*, **491**, 1600. (arXiv:1906.02476)
18. Lyman-alpha Transmission Properties of the Intergalactic Medium in the CoDa II Simulation, Gronke, M., Ocvirk, P., ..., **Shapiro, P. R.**, Yepes, G., 2020, *MNRAS*, **508**, 3679. (arXiv:2004.14496)
19. Large-Scale Variation in Reionization History Caused by Baryon-Dark Matter Streaming Velocity, Park, H., **Shapiro, P. R.**, Ahn, K., Yoshida, N., Hirano, S., 2020, *ApJ*, **908**, 96. (arXiv:2010.12374)
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21. The Impact of Inhomogeneous Subgrid Clumping on Cosmic Reionization - II. Modelling Stochasticity, Bianco, M., Iliev, I. T., Ahn, K., Giri, S., Mao, Y., Park, H., **Shapiro, P. R.**, 2021, *MNRAS*, **504**, 2443. (arXiv:2101.01712)
22. Core-Envelope Haloes in Scalar Field Dark Matter with Repulsive Self-Interaction: Fluid Dynamics Beyond the de Broglie Wavelength, Dawoodbhoy, T., **Shapiro, P. R.**, Rindler-Daller, T., 2021, *MNRAS*, **506**, 2418. (arXiv:2104.07043)
23. Crucial Factors of Ly-α Transmission in the Reionizing Intergalactic Medium: Infall Motion, HII Bubble Size, and Self-shielded Systems, Park, H., Jung, I., Song, H., Ocvirk, P., **Shapiro, P. R.**, ..., 2021, *ApJ*, **922**, 263. (arXiv:2105.10770)
24. Cosmological Structure Formation in Scalar Field Dark Matter with Repulsive Self-Interaction: *The Incredible Shrinking Jeans Mass*, **Shapiro, P. R.**, Dawoodbhoy, T., Rindler-Daller, T., 2022, *MNRAS*, **507**, 145. (arXiv:2106.13244)

Other Recent Publications (2017 - present only)

1. Bose-Einstein-Condensed Scalar Field Dark Matter and the Gravitational Wave Background from Inflation: New Cosmological Constraints and its Detectability by LIGO, Li, B., **Shapiro, P. R.**, Rindler-Daller, T., 2017, *PRD*, **96**, 063505. (arXiv:1611.07961)
2. Core-Halo Mass Relation in Scalar Field Dark Matter Models and its Consequences for the Formation of Supermassive Black Holes, Padilla, L. E., Rindler-Daller, T., **Shapiro, P. R.**, Matos, T., Vázquez, J. A., 2021, *PRD*, **103**, 063012. (arXiv:2010.12716)
3. Angular Momentum and the Absence of Vortices in the Cores of Fuzzy Dark Matter Haloes, Schobesberger, S., Rindler-Daller, T., **Shapiro, P. R.**, 2021, *MNRAS*, **505**, 802. (arXiv:2101.04958)
4. Precision cosmology and the stiff-amplified gravitational-wave background from inflation: NANOGrav, Advanced LIGO-Virgo and the Hubble tension, Li, H., **Shapiro, P. R.** 2021, *JCAP*, **2021**, 24. (arXiv:2107:12229)

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RESEARCH INTEREST

Theoretical & Computational Astrophysics, Especially Cosmology
simulating galaxy formation & evolution;
numerical cosmological simulation of epoch of reionization;
role of dark matter models in structure formation;
usage of machine learning in simulation analysis

EDUCATION

Ph.D. in Astronomy, University of Texas at Austin — 09/2021 - present
Supervisor: Prof. Paul Shapiro

B.Sc. in Physics, Seoul National University (SNU) — 03/2014 - 08/2021
B.Eng. in Electrical and Computer Engineering, SNU — 03/2014 - 08/2021

RESEARCH EXPERIENCE

Research Associate, Computational Cosmology Group, SNU — 09/2019 - 08/2021
Supervisor: Prof. Ji-hoon Kim

- Estimating Galaxy Properties from Their Dark Matter Using Machine Learning
  - Applied trained machine to the cosmological simulation halo catalog (IllustrisTNG)
  - Compared 2PCF in IllustrisTNG halo catalog and machine-predicted halo catalog
- High-velocity Galaxy Collision in Large Cosmological Simulation
  - Searched IllustrisTNG catalog for a high-velocity galaxy collision event
- Simulating of Formation of Dark Matter Deficient Galaxies and Star Clusters
  - Runned a suite of 1.25 pc-resolution galaxy collision simulations with different parameters
  - Resolved the formation process of dark matter deficient galaxies and massive star clusters

Research Associate, AGN Research Group, SNU — 09/2020 - 02/2021
Supervisor: Prof. Jong-Hak Woo

- Calibrated and Applied Novel Method of Measuring SFR in AGNs
  - Tested Oxygen lines as SFR indicator by analyzing SDSS spectroscopy data and IR surveys
  - Investigated correlation between gas outflow strength from AGNs and SFR of host galaxies

AWARDED FELLOWSHIPS & SCHOLARSHIPS

Dean's Excellence Fellowship, University of Texas at Austin — 09/2021 - 08/2022
Presidential Science Scholarship, Korea Student Aid Foundation — 03/2014 - 08/2020

PUBLICATIONS

1. Shin, E. -j., Jung, M., Kwon, G., Kim, J. -h, **Lee, J.**, Jo, Y., & Oh, B. K., "Dark Matter Deficient Galaxies Produced Via High-velocity Galaxy Collisions In High-resolution Numerical Simulations", *ApJ 899 (2020) 25*, arXiv:2007.09889
2. **Lee, J.**, Shin, E. -j., & Kim, J. -h., "Dark Matter Deficient Galaxies And Their Member Star Clusters Form Simultaneously During High-velocity Galaxy Collisions In 1.25 pc Resolution Simulations", *ApJL 917 (2021) L15*, arXiv:2108.01102

TALKS & PRESENTATIONS

Galaxy Evolution Workshop 2021, ASIAA	02/2022
Numerical Galaxy Formation Mini-Workshop, SNU	01/2022
SAZERAC-SIPS Early Galaxy Formation Near and Far	12/2021
The 1st KIAA Forum on Gas in Galaxies for Early Career Scientists	10/2021
UT Austin Extragalactic/Cosmology Seminar	09/2021
AGORA WORKSHOP 2021	08/2021

COMPUTING SKILLS & EXPERIENCES

Languages: Python, C, C++, LaTex (skilled);
Fortran, MATLAB, Mathematica, html, Markdown (familiar);
IDL, RISC-V assembly language (basic)
Astrophysical Simulation Codes: Enzo, Gadget, DICE, yt
Machine Learning: PyTorch, TensorFlow (familiar)
High performance computing experience:

- Local cluster of Computational Cosmology Group, SNU (CentOS)
- Nurion, Korea Institute of Science and Technology Information (CentOS),
- Frontera, Texas Advanced Computing Center (CentOS),
- Stampede2, Texas Advanced Computing Center (Red Hat),

CONFERENCES and EXTERNAL SUMMER/WINTER SCHOOLS

2021 Astrophysics Summer School, Korea Institute for Advanced Study (KIAS)	07/2021
Numerical Galaxy Formation Mini-Workshop, SNU	01/2021
SLAC Summer Institute 2020	08/2020
2020 Astrophysics Summer School, KIAS	07/2020
Numerical Galaxy Formation Mini-Workshop, SNU	02/2020
2019 KIAS-SNU Winter Camp on Theoretical Physics, KIAS	01/2020
Nishina School 2019, RIKEN Nishina Center for Accelerator-Based Science	08/2019

OUTREACH & TEACHING EXPERIENCES

Korea Student Aid Foundation Science Teaching Service	01/2015 - 02/2015
Habitat for Humanity in Cebu, Phillippines	02/2016
Military Service at Korean Air Force 5th Air Mobility Wing	05/2017 - 04/2019

OTHER SKILLS

Languages: Korean (native), English, Japanese (fluent)

CURRENT AND PENDING SUPPORT (PI)

CURRENT SUPPORT

Role	Source	Title
PI-Shapiro	NSF XSEDE	"Simulating Cosmic Reionization and Large-Scale Structure: 3-D Radiative Transfer, Gravitational N-body Dynamics and Radiation-Hydrodynamics"

Duration:	4/1/21 – 9/30/22
Computer Allocation:	75,000 node hours, TACC Dell/Intel Knights Landing System ("Stampede2"); 2,564,329 core hours, PSC Bridges-2 Regular Memory ("Bridges2")
Level of effort:	1 Calendar Month = 0.08 WY
XSEDE Contact:	Vicki Halberstadt, vickih@ncsa.uiuc.edu, (217) 244-5709

Role	Source	Title
Co-PI-Shapiro	DOE INCITE	"Simulating Reionization of the Local Universe: Witnessing Our Own Cosmic Dawn"

Duration:	1/1/20 – 02/28/23
Computer Allocation:	620,000 Summit node hours, Oak Ridge National Laboratory, IBM AC922 ("Summit")
Level of effort:	1 Calendar Month = 0.08 WY
Point of Contact:	Dr. Judith C. Hill, hilljc@DOEleadershipcomputing.org, (865) 576-8453

PENDING SUPPORT

Role	Source	Title
PI-Shapiro	NSF	"Testing Scalar Field Dark Matter Models for Cosmology and Structure Formation"

Duration:	7/1/22 – 6/30/25
Total:	\$569,640
Level of effort:	2 Calendar Month = 0.17 WY
Point of Contact:	Dr. Nigel Sharp, nsharp@nsf.gov, (703) 292-4905

Role	Source	Title
PI-Shapiro	NSF	"Simulating Cosmic Dawn: Inhomogeneous Reionization and Galaxy Formation on Scales Large and Small"

Duration:	7/1/22 – 6/30/25
Total:	\$569,640
Level of effort:	2 Calendar Month = 0.17 WY
Point of Contact:	Dr. Nigel Sharp, nsharp@nsf.gov, (703) 292-4905

CURRENT AND PENDING SUPPORT (FI)

No C&P funding to report

MENTORING PLAN

PI: Prof. Paul Shapiro
FI: Joohyun Lee

This mentoring plan describes the mentoring opportunity and support the FI, Joohyun Lee, will receive from the PI and supervisor, Prof. Paul Shapiro as well as the department. The PI and the Department of Astronomy at UT Austin are committed to ensuring that graduate students are provided with support for their success and well-being. The FI will be assisted with various formal and informal opportunities for scientific and career development throughout and beyond the period of this FINESST support. Through a program of structured mentoring activities, the FI will be provided with assistance to improve the skills, knowledge, and experience necessary to succeed in his chosen career path. To accomplish this goal, the PI and FI have developed this mentoring plan. Specific elements of the mentoring plan include:

- **Regular Research Meetings**: The FI and PI will meet at least once a week to discuss the research outlined in this proposal, and more frequently during peak activity. To ensure research milestones are achieved, these meetings will include clear discussions of research progress while motivating healthy work habits. The FI, PI and his other group members will also participate in the weekly video conference of his international Cosmic Dawn (CoDa) collaboration to discuss research progress. By attending weekly group meetings, the FI will also interact with other group members including postdocs, graduate students, and undergraduate students to engage in discussions of ongoing research and current literature. Finally, regular video conferences will be held to discuss research progress with proposal collaborator, Prof. Ji-hoon Kim, at Seoul National University in Korea.

- **Regular Article Reading Meetings**: The FI will attend the weekly Texas Cosmology Center (TCC) journal club hosted by the PI, a gathering of graduate students, postdocs, visitors, and faculty from both the Astronomy and Physics Departments, to discuss the latest preprints, followed by continued interaction at the PI's Friday Astrophysics Lunch at the UT Campus Club. The FI will also attend the weekly Exgal arXiv/Coffee Hour hosted within the Astronomy Department.

- **Presentation Opportunities**: The FI will have a variety of opportunities to improve his communication and presentation skills in various settings. Students regularly give research presentations and practice job talks at the weekly Graduate Student-Postdoc seminar and in our topical weekly research seminars (e.g. Extragalactic/Cosmology seminar). At the TCC journal club and Exgal arXiv/Coffee Hour paper journal club, the FI will present at least more than once a semester in each seminar about interesting papers that cover a wide range of topic including observational/theoretical works on cosmology, extragalactic astronomy, and galaxy formation and evolution.

- **Publications and Writing Support**: The work supported by FINESST will lead to a number of publications. The FI will receive appropriate guidance and training in the preparation of manuscripts for scientific journals.

- **Assessment and Individual Development Plan**: In order to ensure the communication between advisor and student on needs, goals, plans, and progress, the department has implemented an annual progress report assessed by the FI's Ph.D. committee. This document is drafted with joint input from both the advisor and student, and allows them to have an aligned vision on short and long term research, academic, and writing goals.

- **Teaching and Mentorship Training**: The Texas Institute for Discovery Education in Science (TIDES) in the College of Natural Sciences at UT Austin hosts regular workshops open to all educators, and invites participation from graduate students. The Department of Astronomy conducts two summer undergraduate research programs – the Texas Undergraduate Research Experience for Under-represented Students (TAURUS) and an NSF-funded REU program. Graduate students routinely join postdocs in planning and participate in a seminar series for these visiting students. They also serve as graduate student mentors for visiting undergraduate students.

- **Access to Supercomputing Facilities**: Large-scale, massively-paralleled, numerical computation is a crucial element for this project. Students and researchers at UT Austin have access to supercomputing allocation through the Texas Advanced Computing Center (TACC). The FI will be provided with additional allocation resources at TACC and the Pittsburgh Supercomputing Center (PSC) through the PI's NSF-funded XSEDE projects, and at Oak Ridge Leadership Computing Facilities (OLCF) through the PI's DOE INCITE grant.

- **Travel Support**: Travel funds are available through the department and UT Graduate School, and can supplement and help subsidize travel funding provided by research advisors. The FI will attend and present his work at regional, national, and international conferences during the period of this FINESST support.

- **Extended Mentorship**: The FI has opportunities for additional mentorship by and interaction with the many long-time collaborators of the PI throughout the US and the world with whom he is in regular communication. The FI will have frequent opportunities to learn to interact and consult with these PI collaborators, including their students and postdocs, as well, often gaining their assistance in areas such as computation and data management, as well as in more general research and academic topics. Through these relationships with the collaborators, he will have opportunities to collaborate with experts in diverse disciplinary areas.

- **Additional Resources**: The Association of Women in Astronomy Research and Education (AWARE) and Equity & Inclusion groups meet regularly for discussions. Both groups are co-led by graduate students. In addition, students can also directly contribute to department affairs as representatives in graduate admissions committees, faculty hiring processes, and faculty meetings.

BUDGET AND NARRATIVE

	Year 1 09/01/2022 –08/31/2023	Year 2 09/01/2023 –08/31/2024	Year 3 09/01/2024 –08/31/2025
E. Participant/Trainee Support Costs			
1. Tuition/Fees/Health Insurance			
a) Tuition/Fees (12 months/yr)	\$12,000	\$12,500	\$13,000
b) Health Insurance	\$ 4,000	\$ 2,500	\$ 1,000
Subtotal (E.1.a + E.1.b)	\$16,000	\$15,000	\$14,000
2. **Stipend** (12 months/yr)	\$34,000	\$35,000	\$36,000
Budget Total (E.1 + E.2)	\$50,000	\$50,000	\$50,000